

Sudip Bhattarai

BSc CSIT Student | Aspiring Software Developer

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PROFESSIONAL SUMMARY

BSc CSIT student at Tribhuvan University (4th semester) who builds projects in C, C++ and JavaScript, including a canvas-based traffic simulation and a C++/OpenGL paint application. Uses Git and GitHub for version control and is studying data structures, operating systems, computer networks and artificial intelligence. Seeking a software development internship or junior role, with interests in backend development, databases and AI.

TECHNICAL SKILLS

Languages	C, C++, Python (learning)
Web	HTML, CSS, JavaScript (basic)
Graphics & APIs	OpenGL / GLUT (C++), HTML5 Canvas
Tools	Git, GitHub, VS Code
Coursework	Data Structures, Operating Systems, Computer Networks, Artificial Intelligence

SELECTED PROJECTS

TrafficAgent – Traffic Intersection Simulation — github.com/meSudip028/TrafficAgent
JavaScript (ES modules) · HTML5 Canvas · CSS — Live demo: mesudip028.github.io/TrafficAgent

- Built a four-way intersection simulation in JavaScript and HTML5 Canvas, using layered canvases and a *requestAnimationFrame* loop with delta-time updates.
- Implemented a fixed-cycle signal controller (10 s green, 3 s yellow) and a spawner that randomly adds cars, buses, trucks and motorcycles to two lanes per direction when the spawn area is clear.
- Programmed rule-based vehicle behaviour: vehicles brake smoothly for non-green lights and keep a safe gap from the vehicle ahead in their lane.

Mini Paint Application – Computer Graphics Project — github.com/meSudip028/Mini-Paint-Application
C++ · OpenGL · GLUT / freeglut

- Developed an interactive paint program in C++ and OpenGL with line, circle and fill tools, a colour palette, undo and clear-canvas controls.
- Implemented the DDA line algorithm, the Midpoint Circle algorithm and an iterative 4-connected flood fill (explicit stack) over an 800×600 pixel buffer.
- Stored drawn line and circle shapes in a vector to support undo and redraw.

C++ Calculator – Menu-Driven Console Application — github.com/meSudip028/CPP-CALCULATOR
C++ · <cmath>

- Developed a console calculator with 13 operations: arithmetic, exponent, square root, base-10 logarithm, and trigonometric and inverse trigonometric functions with angles in degrees.
- Added input checks for division by zero, negative square-root input, non-positive logarithm input, out-of-range arcsine/arccosine input and invalid menu choices.

EDUCATION

Bachelor of Science in Computer Science and Information Technology (BSc CSIT) — Oct 2024 – Present
Tribhuvan University, Bhaktapur Multiple Campus, Bhaktapur, Nepal · Currently in 4th semester

Higher Secondary Education (Grade 12) — GPA: 3.64 / 4.00

Secondary Education (Grade 10) — GPA: 3.75 / 4.00